Biochemistry Cume

April 3, 2009

A competitive inhibitor can bind to free enzyme but not to the ES complex.

Derive equations showing the fraction of total enzyme present as free E, in the ES complex, and in the EI complex as a function of Km, Ki, substrate concentration (S) and inhibitor concentration (I).

f_E = (E)/(Etotal)

f_{ES} = (ES)/(Etotal)

f_{EI} = (EI)/(Etotal)

(**25 points** for *each* derived equation)

One way to examine whether the equations you have derived are reasonable is to show that the sum of the three fractions adds up to 1.

The enzyme under consideration has a Km value of 10 mM and a Ki of 5 mM. Vmax is 100 units/min. Assuming the concentrations of S and I are both 10 mM, are the equations you derived reasonable?
(**25 points**)